Roll No. | | | | | | | |

ODD End Semester Examination December 2025

Name of the Program: M. Tech **Semester: I**

Name of the Course: Electric Vehicle Technology **Course Code: MEV 105**

Time: 3 Hrs Maximum Marks: 100

Note:-

(i) All questions are compulsory..

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	**(20marks)**	**CO1**
(a)	What is the importance of regenerative breaking in electric vehicle (EV)? Define the sub systems and the essential components of an EV with a suitable block diagram.	
(b)	Differentiate between Series, Parallel, and Series-Parallel hybrid electric vehicle (HEV) with the help of a suitable block diagram.	
(c)	Define and explain any ten performance metrics of an EV. What are the key parameters which make EVs better than internal combustion (IC) engine vehicles?	
Q2	**(20marks)**	**CO2**
(a)	Calculate the total propulsion power required for an EV if the following details are given: Kerb weight = 500 kg, payload = 300 kg, coefficient of rolling resistance = 0.02, air density = 1.25 kg/m^3, frontal area = 2.34 m^2, drag coefficient = 0.33, speed = 80 kmph, slope angle (degree) = 11^0, speed during hill climbing = 80 kmph (assumed).	
(b)	State Newton's Second Law of motion with the appropriate equation. Define and explain different type of forces exist during the motion of the vehicle.	
(c)	What do you mean by drive cycle? What are the advantages of drive cycle? Explain different types of drive cycle.	
Q3	**(20marks)**	**CO3**
(a)	What is the importance of electric transmission in an EV? Differentiate between single-speed and multi-speed transmission. What are the advantages of multi-speed transmission?	
(b)	Explain powertrain layouts on the basis of single motor, dual motor, and in-wheel motor with suitable diagram.	
(c)	What is the importance of thermal management in EV? Describe the key components of the thermal management system required in an EV. Describe any two cooling methods?	
Q4	**(20marks)**	**CO4**
(a)	What is the concept of fuel cell in EV? Explain the working of solid oxide fuel cell (SOFC) and polymer electrolyte membrane fuel cell (PEMFC) with the help of a suitable working diagram.	
(b)	Define the following terms: (i) Battery Swapping Technology (ii) Specific Energy and Energy Density of Battery	

	(iii) State of charge (SOC) and Depth of discharge (DOD) (iv) Ultra-capacitor (v) Cell, modules, and packs of a battery	
(c)	Give classification of Electrochemical Energy Storage sources. What are the advantages and disadvantages of Li-ion batteries?	
Q5	**(20 Marsk)**	CO5
(a)	Differentiate between Level 1, Level 2, and Level 3 charging systems for an EV.	
(b)	What do you understand by V2G (Vehicle to Grid) concept? What are the possible benefits and drawbacks of V2G concept?	
(c)	What are the opportunities and challenges of integrating renewable energy sources with EV charging stations?	